MATH 122 — Calculus II · Exam 2: Techniques of Integration

Name: _____ **Date:** _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
- Points: Each question is worth 10 points.
- Formula Sheet: Permitted (formulas only; no worked examples).
- Calculators: Basic, non-graphing calculators allowed.

Part I: Integration by Parts

1. (10 points) Evaluate

$$\int x^{20} \ln x \, dx$$

2. (10 points) Evaluate

$$\int 3x^3 e^x \, dx$$

3. (10 points) Evaluate

$$\int \cot^{-1}(x)\,dx$$

4. (10 points) Evaluate the definite integral

$$\int_0^{\pi} x \sin x\,dx$$

Part II: Trigonometric Integrals

5. (10 points) Evaluate

$$\int \sin^7 x \, dx$$

6. (10 points) Evaluate

$$\int \sin^3 x \cos^2 x \, dx$$

7. (10 points) Evaluate

$$\int \sin 10x \cos 3x \, dx$$

8. (10 points) Evaluate

$$\int 4 \sin 3x \cos 5x \sin 7x \, dx$$

Part III: Partial Fractions

9. (10 points) Evaluate

$$\int \frac{5}{(x-2)(x+3)}\,dx$$

10. (10 points) Evaluate

$$\int \frac{4}{x^2(x-2)}\,dx$$